

Project Interim Report
On
**Online Course Management System
(Using Core Java & OOP Concepts)**

RU-100-01-00012
Object Oriented Programming with Java

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1.INTRODUCTION

The Online Course Management System is a console-based software application developed using Core Java and Object-Oriented Programming concepts. The system is designed to manage course information, student enrollment, and progress tracking efficiently.

In modern educational environments, managing student records and course information manually can be time-consuming and prone to errors. This project provides a simple solution that automates these processes and demonstrates the practical implementation of Java programming concepts.

The project focuses on object interaction, data management, and modular programming while providing an easy-to-use interface for managing educational activities.

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2.PROJECT OBJECTIVES

- **Develop a course management application using Java.**
- **Apply Object-Oriented Programming principles.**
- **Manage student and course information.**
- **Implement student enrollment functionality.**
- **Track student learning progress.**
- **Improve programming and software development skills.**

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3.PROBLEM STATEMENT

Educational institutions require efficient systems for handling student registrations and course management. Manual methods often lead to data duplication, errors, and difficulty in tracking student progress.

The Online Course Management System addresses these challenges by providing a software solution that stores student and course information, manages enrollments, and displays progress details effectively.

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4.SYSTEM DESIGN

The project consists of the following classes:

Course Class:

- **Course Name**
- **Course ID**
- **Duration**

Student Class:

- **Student Name**
- **Student ID**
- **Progress Percentage**
- **Enrolled Course**

Main Class:

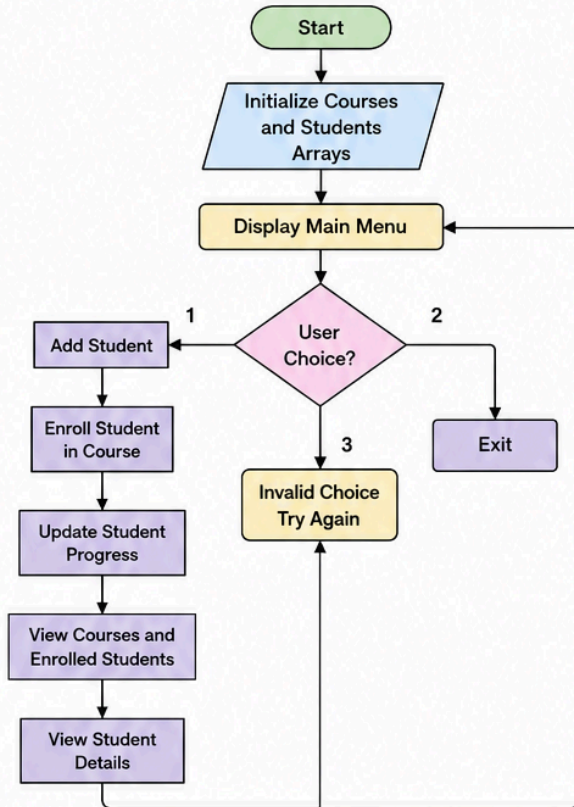
- **Creates objects**
- **Stores data in arrays**
- **Handles enrollment**
- **Displays reports**

5. FLOWCHART

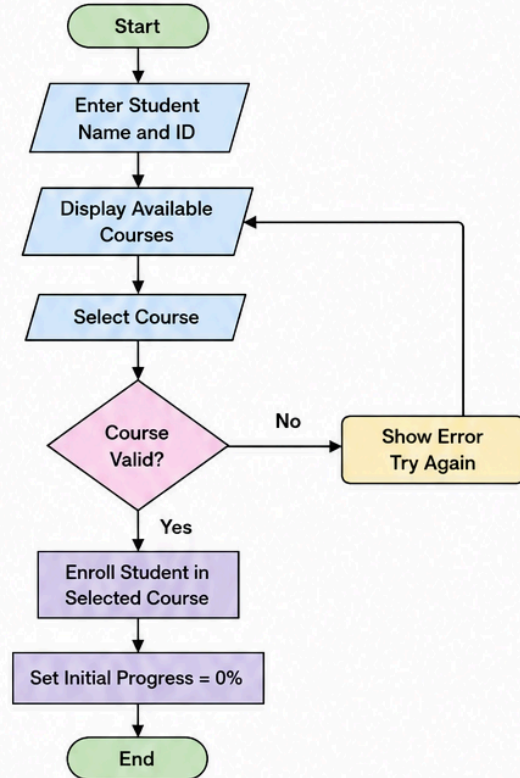
FLOWCHARTS

The following flowcharts represent the working and logic of the Online Course Management System.

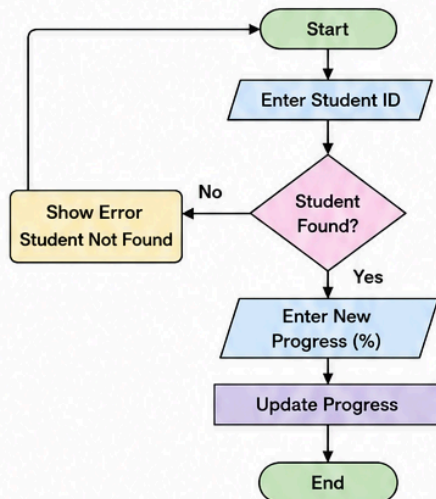
1. MAIN FLOWCHART



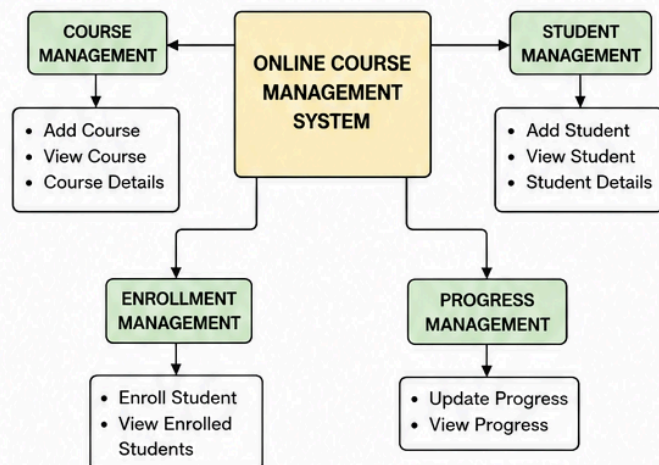
2. STUDENT ENROLLMENT FLOWCHART



3. PROGRESS UPDATE FLOWCHART



4. SYSTEM OVERVIEW FLOWCHART



FLOWCHART DESCRIPTION

- **Main Flowchart:** Shows the overall flow of the system starting from initialization to menu-driven operations.
- **Student Enrollment Flowchart:** Explains the step-by-step process of enrolling a student in a course.
- **Progress Update Flowchart:** Shows how progress is updated for a particular student.
- **System Overview Flowchart:** Represents the structure and modules of the system.

Caption:

Figure 1: Flowchart of Online Course Management System

Description:

The flowchart represents the overall working process of the system. The program starts by creating course and student objects. Students are enrolled in courses, progress is updated, and information is displayed through the console interface.

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6.METHODOLOGY

Step 1:

Create the Course class and initialize course details.

Step 2:

Create the Student class and initialize student details.

Step 3:

Implement constructors for object initialization.

Step 4:

Create methods for enrollment and progress tracking.

Step 5:

Store objects using arrays.

Step 6:

Display complete information through loops.

Step 7:

Test the application and verify outputs.

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7.FEATURES OF THE SYSTEM

- **Course Management**
- **Student Management**
- **Enrollment Tracking**
- **Progress Monitoring**
- **Data Display**
- **Array-Based Storage**

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8.TECHNOLOGIES USED

Programming Language : Java

IDE : Visual Studio Code

Compiler : JDK

Platform : Windows

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9.OOP CONCEPTS USED

Classes and Objects:

Used to represent students and courses.

Encapsulation:

Data is organized within classes.

Constructors:

Used for object initialization.

Methods:

Perform enrollment and progress updates.

Arrays:

Store collections of students and courses.

Object Interaction:

Students interact with Course objects.

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10. SOURCE CODE

File Name:

OnlineCourseManagementSystem.java

(Paste Complete Source Code Here)

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11. OUTPUT SCREENSHOTS

11.1 Source Code Screenshot

(Insert VS Code Source Code Screenshot)

11.2 Program Output Screenshot

(Insert Output Screenshot)

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12. RESULTS

The project was successfully implemented and executed. Students were enrolled in courses and

their progress was updated and displayed correctly.

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13. ADVANTAGES

- **Easy to understand.**
- **Demonstrates OOP concepts clearly.**
- **Efficient course and student management.**
- **Extendable for future enhancements.**

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14. LIMITATIONS

- **Fixed-size arrays.**
- **No database support.**
- **Console-based interface only.**
- **No user authentication.**

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15. FUTURE ENHANCEMENTS

- Database connectivity using MySQL.
- GUI using Java Swing.
- Dynamic storage using ArrayList.
- Student search functionality.
- Multiple course enrollment.
- Online assessment module.

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16. CONCLUSION

The Online Course Management System successfully demonstrates the implementation of Core Java and Object-Oriented Programming concepts. The project manages students, courses, enrollments, and progress tracking in an efficient manner.

The project enhanced understanding of Java classes, objects, constructors, methods, arrays, and object interaction. It provides a strong foundation for developing larger educational management systems in the future.

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17. REFERENCES

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19. Java: The Complete Reference by Herbert Schildt

20. VS Code Documentation

21. Java Programming Tutorials

22. Course Study Material

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